

1 Introduction

The proliferation of generative artificial intelligence (AI) has sparked a global debate about its potential impact on the labor market. This discourse spans utopian predictions of enhanced productivity, dystopian fears of widespread job displacement, and skeptical views that AI will have minimal effects on employment or productivity. Historically, technologies have affected different tasks, occupations, and industries in different ways, replacing work in some, augmenting others, and transforming still others. These heterogeneous effects suggest that there may be “canaries in the coal mine” which are harbingers of more widespread effects of AI.

Recent discussion coincides with rapid improvements in AI capabilities and adoption. From 2023 to 2024, AI systems improved from solving 4.4% of coding problems to 71.7% on SWE-Bench, a widely used benchmark for software engineering (Maslej et al., 2025). AI has also improved in areas like language understanding, subject knowledge, and reasoning. A recent paper found that current systems could match or outperform up to 47 percent of industry professionals on a pre-defined benchmark of economically valuable tasks (Patwardhan et al., 2025). At the same time, AI systems are increasingly widely adopted. Hartley et al. (2025) find that LLM adoption at work among U.S. survey respondents above age 18 reached 46% by June/July 2025.¹

Given the better capabilities and widespread adoption, a central concern, amplified in recent headlines, is whether AI is beginning to supplant human labor, particularly for younger, entry-level workers in highly exposed professions like software engineering and customer service.² For instance, in May of 2025, Dario Amodei, co-founder and CEO of Anthropic predicted that AI could wipe out roughly 50 percent of all entry-level white-collar jobs within five years (Morris, 2025).

Despite the intensity of this debate, empirical evidence has struggled to keep pace with technological advancement, leaving many fundamental questions unanswered. This paper confronts this empirical gap by leveraging a large-scale, high-frequency administrative dataset from ADP, the largest payroll software provider in the United States. Our sample consists of monthly, individual-level payroll records through September 2025, encompassing millions of workers across tens of thousands of firms. This rich panel structure allows us to track employment dynamics with a high

¹Similarly, Bick et al. (2024) found that in late 2024, nearly 40% of the U.S. population age 18-64 reported using generative AI, with 23% using it for work weekly and 9% daily.

²Improved productivity of workers in an occupation could lead to either reduced or increased employment, depending on, among other things, how elastic demand is for the output of those workers.

as Eloundou et al. (2024); Felten et al. (2023); Gmyrek et al. (2023); Handa et al. (2025), and Tomlinson et al. (2025) adapted this approach for Generative AI, forming the basis for exposure metrics used in this analysis. While these studies identify potential disruption, our paper connects these exposure measures to actual employment changes.

Our work broadens insights from studies that find significant effects in more specific settings, such as on online freelance platforms (Hui et al., 2023; Demirci et al., 2025) or within individual firms (Brynjolfsson et al., 2025; Dillon et al., 2025).⁶ We measure labor market changes across occupations spanning the US economy.

In this sense our work complements a small but growing list of papers using economy-wide data to measure AI’s impact. Recent findings have been varied. Jiang et al. (2025) find AI exposure correlates with longer work hours in the U.S.⁷ Hampole et al. (2025) use job postings and LinkedIn profiles from Revelio labs from 2011 to 2023 to find limited employment impacts overall, with growing labor demand at firms offsetting relative declines in demand for exposed occupations. Chandar (2025b) uses data from the CPS to compare employment changes in more and less AI-exposed professions, finding little differential trend overall but noting difficulty with measuring changes for young workers given limited effective sample size.⁸ Johnston and Makridis (2025) find employment increases in state-industry pairs more exposed to AI using Quarterly Census of Employment and Wages (QCEW) data.⁹ These prior papers use data lacking either sufficient granularity or immediacy to reliably study employment changes by AI exposure and age (O’Brien, 2025).¹⁰ In contrast, this paper uses large-scale, close to real-time data to take a step towards resolving the ongoing debate on the employment effects of AI on young workers.

After we first publicly shared our results, work by Hosseini and Lichtinger (2025) and Klein Teeselink (2025) similarly found declines in AI-exposed entry-level employment using LinkedIn data from the

⁶See also Noy and Zhang (2023); Peng et al. (2023); Dell’Acqua et al. (2023).

⁷See also Acemoglu et al. (2022); Bonney et al. (2024); Bick et al. (2024); Hartley et al. (2025); Frank et al. (2025); Chen et al. (2025).

⁸See Dominski and Lee (2025); Gimbel et al. (2025); Eckhardt and Goldschlag (2025), which also use CPS data.

⁹Johnston and Makridis (2025) measure state-industry exposure by taking an average of Eloundou et al. (2024)’s occupational exposure weighted by state-industry employment. Industry-level labor market changes may be distinct from the occupation-level changes studied in this paper if firms make capital investments or become more productive in ways that increases overall labor demand (Hampole et al., 2025).

¹⁰As a comparison, the CPS surveyed between 44,000 and 51,000 employed individuals in total across all age groups in each month since 2021. Between 10,000 and 12,000 of these observations were in the outgoing rotation group and included earnings records. The data in our main analysis sample includes between 250,000 and 350,000 employed individuals in each month *just between the ages of 22 and 25*, all with earnings records.

US and UK. In contrast, the most recent version of Humlum and Vestergaard (2025) found minimal effects on entry-level earnings or hours worked in Denmark.¹¹

3 Data Description

3.1 Payroll Data

This study uses data from ADP, the largest payroll processing firm in America. The company provides payroll services for firms employing over 25 million workers in the US. We use this to track employment changes for workers in occupations measured as more or less exposed to artificial intelligence.

We make several restrictions for our main analysis sample. We include only workers with positive earnings, exclude part-time employees, and subset to people under age 70.¹²

The set of firms using payroll services changes over time as companies join or leave ADP’s platform. We maintain a consistent set of firms by keeping only companies that have employee earnings records for each month from January 2021 through September 2025.

ADP observes job titles for about 70% of workers in its system. We exclude workers without a recorded job title. There are over 7,000 standardized job titles, including “Search engineer optimization specialist,” “Enterprise content management manager,” and “Plant documentation control specialist.” The company’s internal research team maps each of these job titles to a 2010 Standard Occupational Classification (SOC) code, additionally using information such as the job description, industry, location, and other relevant data. We use these estimated SOC codes to merge our data to occupational AI exposure measures.

After these restrictions we have records on 3.5 and 5 million workers each month for our main analysis sample, though we consider robustness to alternative analyses such as allowing for firms to enter and leave the sample.

While the ADP data include millions of workers in each month, the distribution of firms using

¹¹An important direction for future work is to reconcile differences among these recent studies, assessing the extent they arise from differences in labor market institutions, measurement of AI adoption, statistical methodology, or other factors.

¹²While we observe the year of birth for each worker, for privacy reasons we do not observe the exact date of birth. We impute month of birth from the distribution of birth months in the United States using data from the Center for Disease Control and Prevention.